# EVPN Solutions/Applications for DC & SP Segments

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# EVPN History

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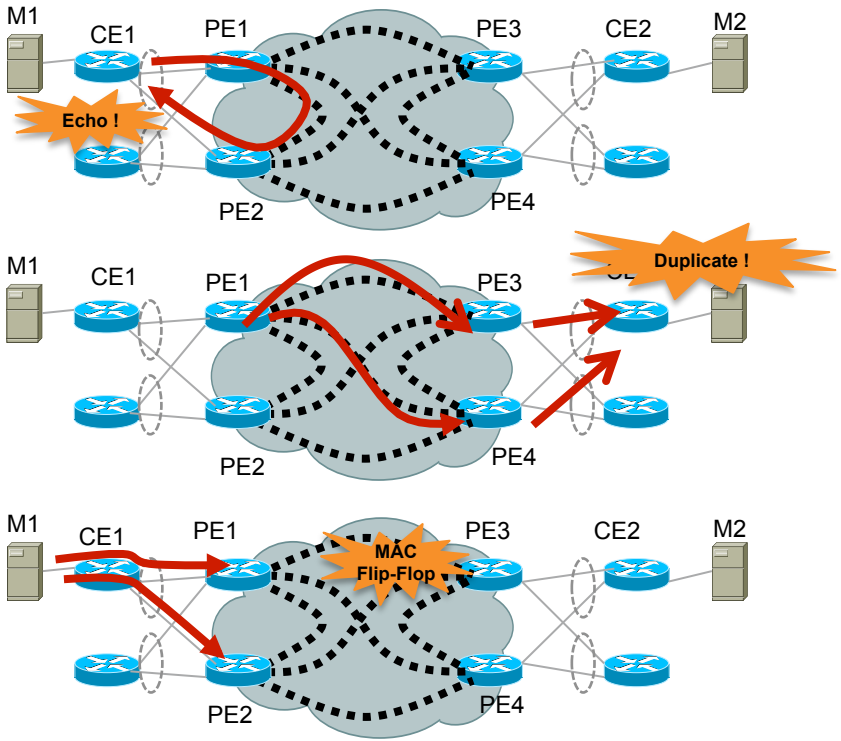
# Challenges with Legacy VPN (VPLS)

- Next generation VPN solution need to have
  - Extensive multi-homing capabilities
  - Per-flow load-balancing in the core and access with DF election and loop prevention
  - Fast convergence upon failure
- Existing VPLS solutions do not offer an All-Active per-flow redundancy
- Looping of Traffic Flooded from PE
- Duplicate Frames from Floods from the Core
- MAC Flip-Flopping over Pseudowire
  - E.g. Port-Channel Load-Balancing does not produce a consistent hash-value for a frame with the same source MAC (e.g. non MAC based Hash-Schemes)





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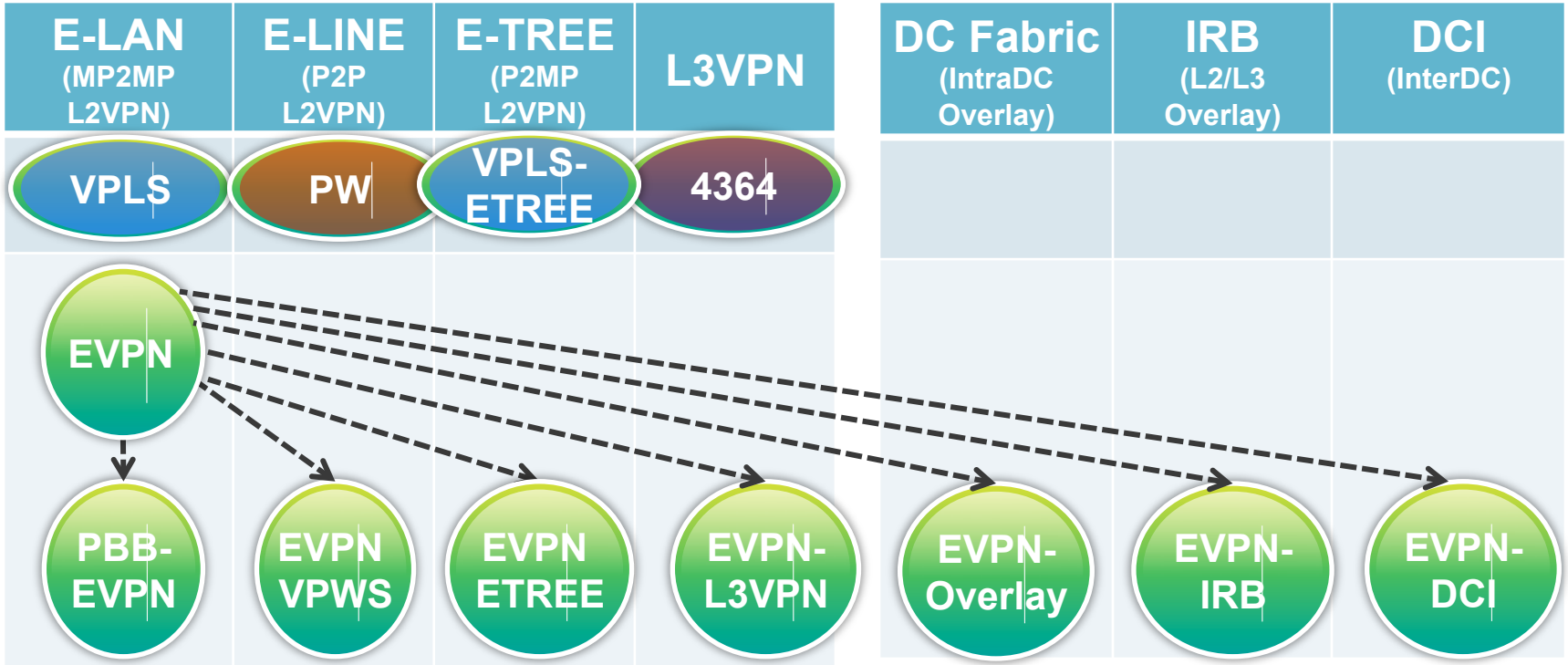
# What is EVPN ?

EVPN is next generation all-in-one VPN solution





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# What's the big deal ?

Not only a single technology/solution does the job of many existing solutions but it does it better!

| Service | Additional Capabilities |
|---|---|
| E-LAN | • Provides All-Active multi-homing<br>• Prevents loop for both all-active & single-active even in transient state<br>• Ability to do per-flow LB & DF |
| E-Line | • Both single-segment & multi-segment support<br>• Discovery & signaling via single protocol – BGP<br>• All-active & single-active redundancy support |
| E-TREE | • Ingress filtering for traffic destined to egress leaf sites |


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# What's the big deal ? – Cont.

## New Applications & Solutions

| Service | Features |
|---|---|
| DC-Fabric (IntraDC Overlay) | • Geo-redundancy & VM mobility support<br>• ARP suppression & ARP proxy<br>• Support for different encaps such as VxLAN, NVGRE, MPLS, MPLSoUDP<br>• Extensive multi-homing support<br>• Extensive load-balancing capabilities |
| IRB | • both L2 & L3 (on a per flow) for a given VLAN/tenant<br>• L3 only mode when needed<br>• Inherent support for anycast GW |
| DCI (InterDC) | • Support for any kind of access technology such as EVPN Overlay (VxLAN), Native Ethernet, TRILL, 802.1Qbp, MPLS<br>• Seamless interop with IP-VPN (RFC4364) |


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# SDO Status

| Service | IETF drafts |
|---|---|
| E-LAN | • RFC7209 – draft-ietf-l2vpn-evpn-req<br>• RFC 7432 - draft-ietf-l2vpn-evpn<br>• RFC 7623 - draft-ietf-l2vpn-pbb-evpn<br>• draft-ietf-bess-evpn-vpls-seamless-integ<br>• draft-rbadan-bess-evpn-optimized-ir<br>• drft-mohanty-bess-evpn-df-election<br>• draft-sajassi-bess-evpn-igmp-mld-proxy<br>• Draft-sajassi-bess-pbb-evpn-anycast-ip-tunnel |
| E-Line | • draft-ietf-bess-evpn-vpws<br>• draft-boutros-bess-evpn-service-edge-gw |
| E-TREE | • draft-bess-l2vpn-evpn-etree |


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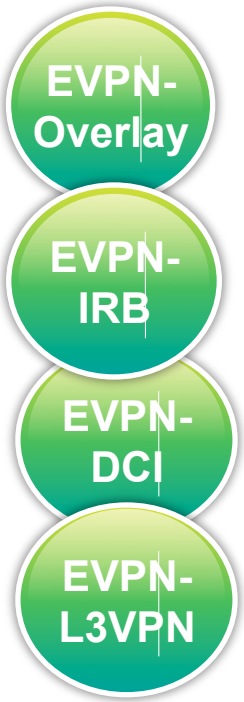
# SDO Status – Cont.




| Service | IETF drafts |
|---|---|
| DC-Fabric (IntraDC Overlay) | • draft-ietf-bess-evpn-overlay |
| IRB | • draft-bess-l2vpn-evpn-inter-subnet-forwarding<br>• draft-bess-l2vpn-evpn-prefix-advertisement |
| DCI (InterDC) | • draft-sajassi-l2vpn-evpn-ipvpn-interop<br>• draft-ietf-l2vpn-trill-evpn<br>• Draft-ietf-l2vpn-spb-evpn<br>• draft-boutros-l2vpn-vxlan-evpn |
| L3VPN | • draft-sajassi-evpn-l3vpn-multihoming |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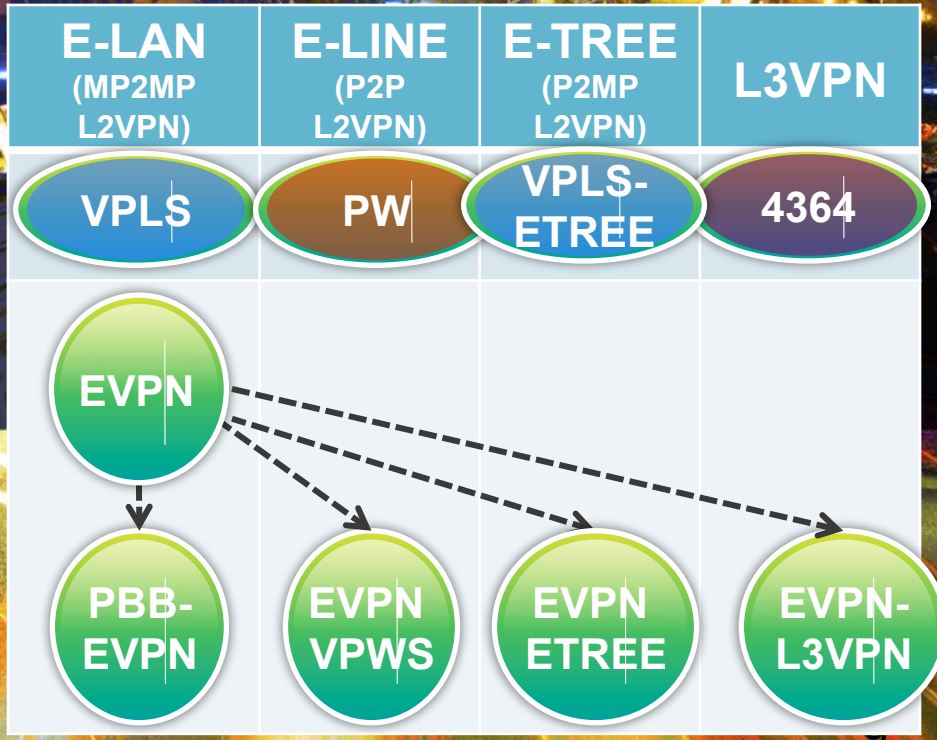
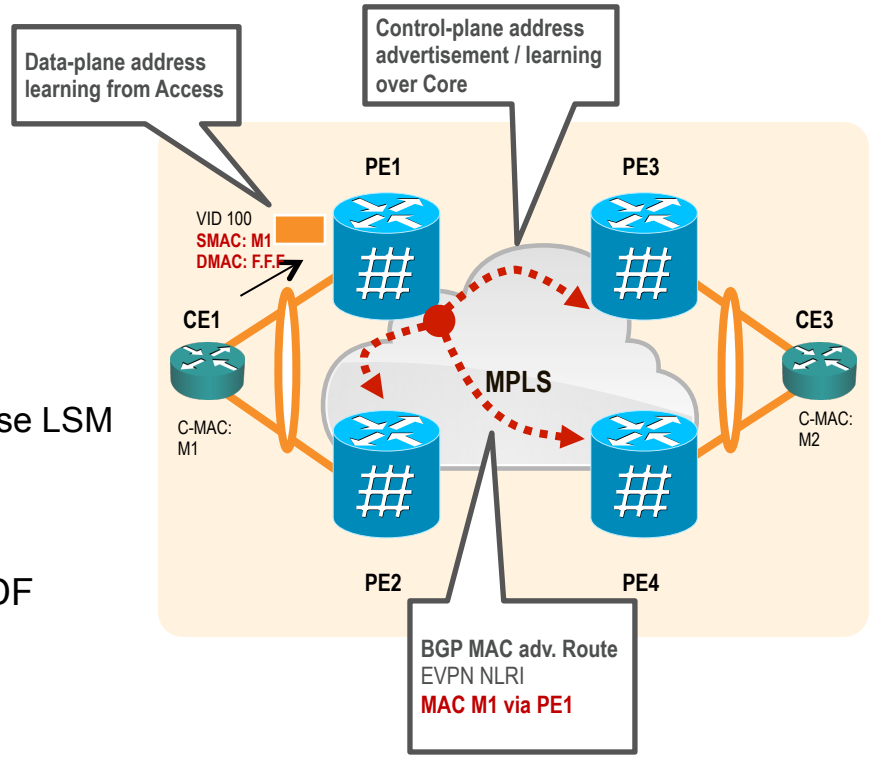
# SP Applications



# EVPN - Highlights

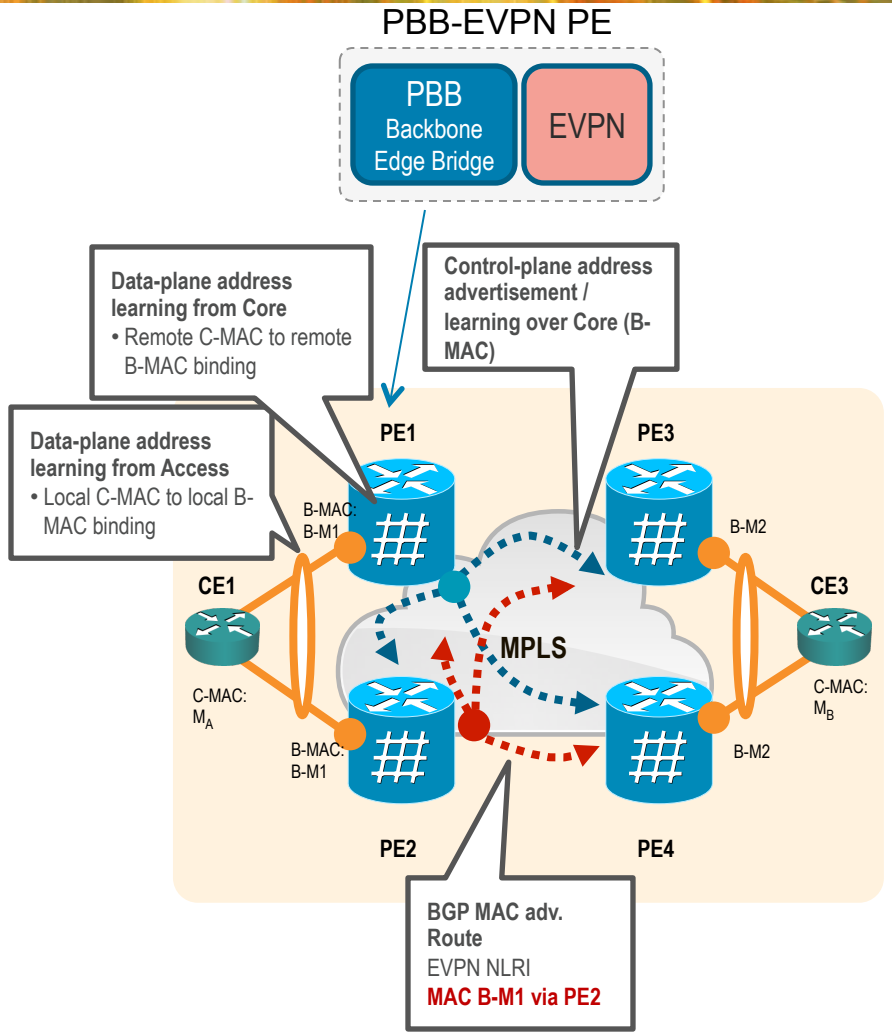
- Next generation solution for Ethernet multipoint (E-LAN) services
- PEs run Multi-Protocol BGP to advertise & learn Customer MAC addresses (C-MACs) over Core
  - Same operational principles of L3VPN
- Learning on PE Access Circuits via data-plane transparent learning
- No pseudowire full-mesh required
  - Unicast: use MP2P tunnels
  - Multicast: use ingress replication over MP2P tunnels or use LSM
- Provides
  - Extensive multi-homing capabilities
  - Per-flow load-balancing in the core and access with DF election and loop prevention
  - Fast convergence upon failure
- Multi-vendor Solution – RFC 7432



# PBB -EVPN

## Highlights

- Next generation solution for Ethernet multipoint (E-LAN) services by combining Provider Backbone Bridging (PBB - IEEE 802.1ah) and Ethernet VPN
- Data-plane learning of local C-MACs and remote C-MAC to B-MAC binding
- PEs run Multi-Protocol BGP to advertise local Backbone MAC addresses (B-MACs) & learn remote B-MACs
  - Takes advantage of PBB encapsulation to simplify BGP control plane operation – faster convergence
  - Lowers BGP resource usage (CPU, memory) on deployed infrastructure (PEs and RRs)
  - Scales for very large number of MACs
- Multi-vendor solution – RFC 7623





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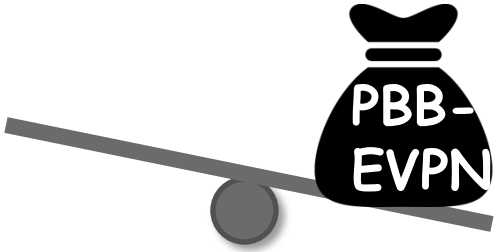
# When to use PBB-EVPN?

- For SP & DCI applications where E2E L2 services needed
- Lower control-plane overhead than EVPN alone
  - PBB-EVPN uses only a sub-set of EVPN routes
  - Simpler and Faster failure convergence for all-active multi-homing scenarios
  - Faster MAC move convergence handled in data-plane
- Lower control-plane scale requirements than EVPN alone
  - BGP MAC advertisements for smaller Backbone MAC (B-MAC) address space
  - Requires less resources (CPU, memory) on deployed infrastructure (PEs / RRs)



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# Comparison of L2VPN Solutions

| Requirement | VPLS | PBB-VPLS | EVPN | PBB-EVPN |
|---|---|---|---|---|
| **Multi-Homing with All-Active Forwarding** | | | | |
| Service Based Load-balancing CE-to-PE | ✔ | ✔ | ✔ | ✔ |
| Flow Based Load-balancing CE-to-PE | ⊘ | ⊘ | ✔ | ✔ |
| Flow Based Load-balancing PE-to-PE | ⊘ | ⊘ | ✔ | ✔ |
| Flow Based Multi-Pathing in the Core | ✔ | ✔ | ✔ | ✔ |
| **Provisioning Simplicity** | | | | |
| Core Auto-Discovery | ✔ | ✔ | ✔ | ✔ |
| Access Auto-Sensing | ⊘ | ⊘ | ✔ | ✔ |
| Redundancy Group Auto-Discovery | ⊘ | ⊘ | ✔ | ✔ |
| Automatic Designated Forwarder election and Service Carving | ⊘ | ⊘ | ✔ | ✔ |
| **Service Interfaces** | | | | |
| Port-Based / VLAN-based / VLAN Bundling | ✔ | ✔ | ✔ | ✔ |
| VLAN-aware Bundling | ⊘ | ⊘ | ✔ | ✔ |
| **Multi-Destination Traffic Forwarding** | | | | |
| Ingress Replication | ✔ | ✔ | ✔ | ✔ |
| LSM with P2MP Tree | ✔ | ✔ | ✔ | ✔ |
| LSM with MP2MP Tree | ⊘ | ⊘ | ✔ | ✔ |

# Comparison of L2VPN Solutions (cont.)

| Requirement | VPLS | PBB-VPLS | EVPN | PBB-EVPN |
|---|---|---|---|---|
| **Fast Convergence** | | | | |
| CE-PE Link Failures / PE Node Failures | ✔ | ✔ | ✔ | ✔ |
| MAC Mobility | ✔ | ✔ | ✔ | ✔ |
| CE-PE Link Failures with Local Repair | ⊘ | ⊘ | ✔ | ✔ |
| **MAC Scalability** | | | | |
| Scale to Millions of C-MAC Addresses | ⊘ | ✔ | ⊘ | ✔ |
| Confinement of C-MAC entries to PE with active flows | ✔ | ✔ | ⊘ | ✔ |
| MAC Summarization | ⊘ | ⊘ | ✔ | ✔ |
| MAC Summarization co-existence with C-MAC Mobility | ⊘ | ⊘ | ⊘ | ✔ |
| **Flexible VPN Policies** | | | | |
| Per C-MAC Forwarding Control Policies | ⊘ | ⊘ | ✔ | ⊘ |
| Per-Segment Forwarding Control Policies | ⊘ | ⊘ | ✔ | ✔ |

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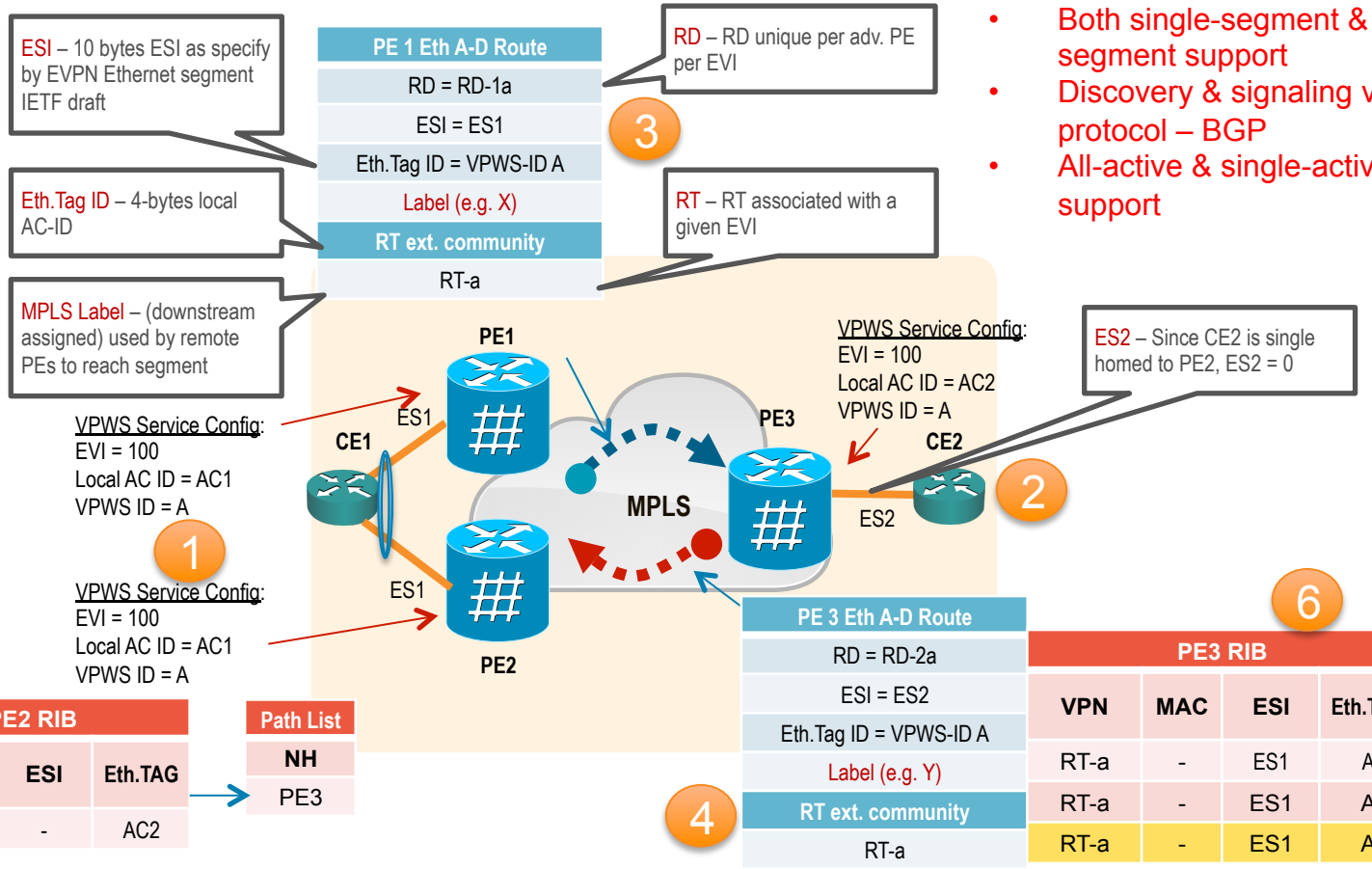
# EVPN VPWS: All-Active Operation




- Both single-segment & multi-segment support
- Discovery & signaling via single protocol – BGP
- All-active & single-active redundancy support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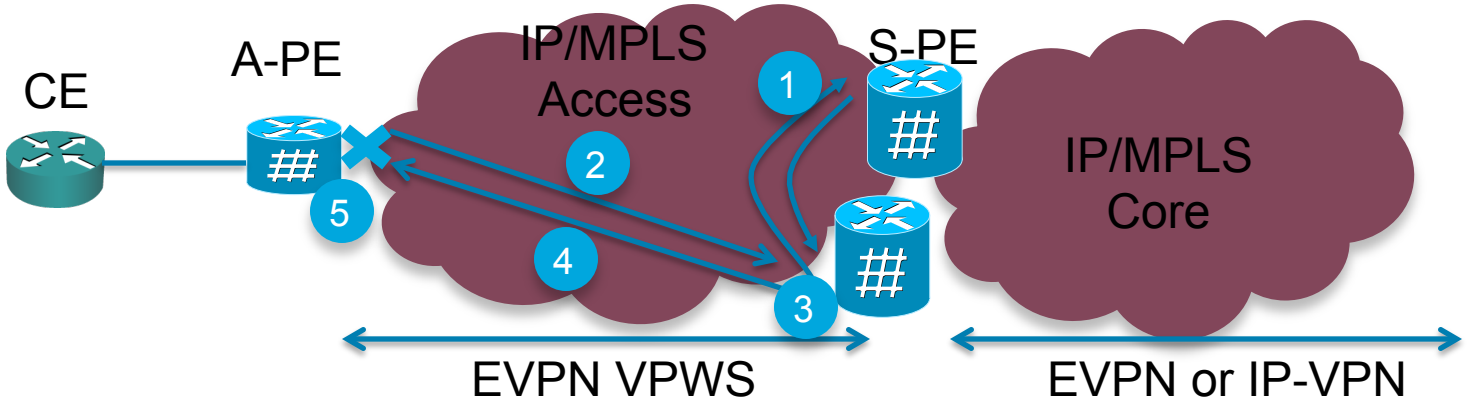
# EVPN-VPWS: Service Edge GW




1. Discovery among S-PEs per EVPN/IP-VPN service needing VPWS
2. Origination of single-side signaling by an A-PE for a given VPWS service (identified by the 24-bit Ethernet tag)
3. DF election by S-PEs for Active/Backup
4. Active & Backup S-PEs responding to the single-sided signaling of step (2)
5. Binding of two halves of the EVC (or IPVC) by the A-PE upon receiving the response in step (3) & setting up backup EVC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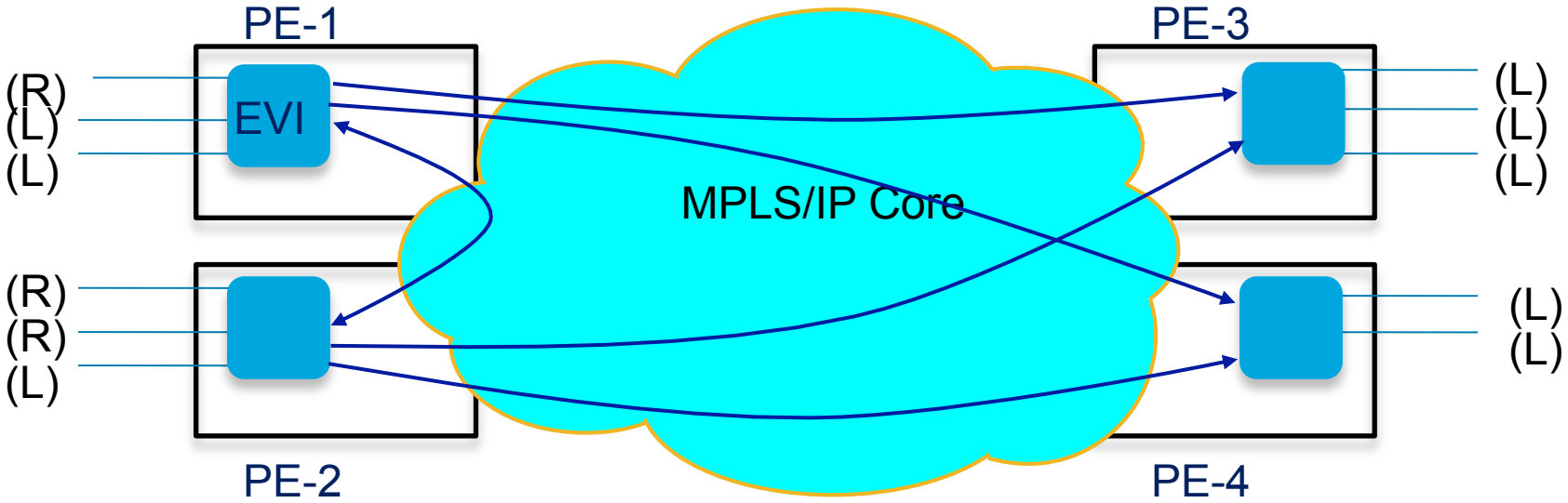
# EVPN-ETREE




EVPN-ETREE provides ingress filtering for traffic destined to egress leaf sites !!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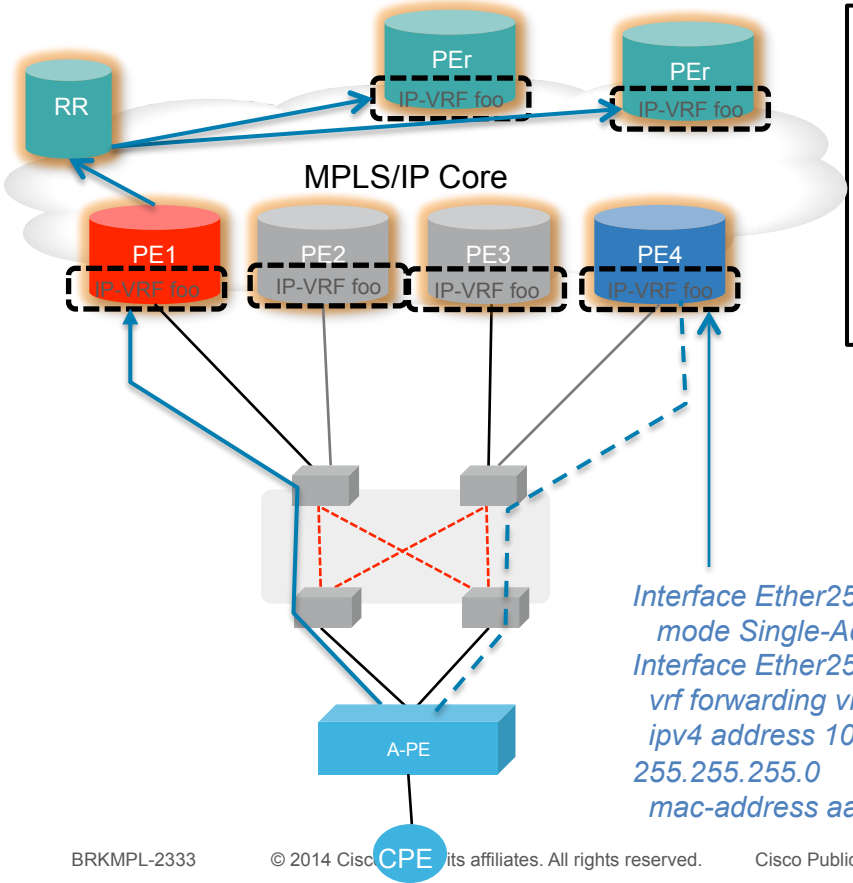
# EVPN-L3VPN




- Req: Single-BGP session from CPE with fast switchover and min. traffic disruption during failure
- Solution has three components:
  1. PIC functionality for fast switchover at PEr devices
  2. Synch up of ARP cache and VRF tables on all PEs in the redundancy group
  3. BGP GR with non-stop forwarding between CPE and PEs in the redundancy group

```
Interface Ether25
  mode Single-Active
Interface Ether25.1
  vrf forwarding vrf-foo
  ipv4 address 10.0.0.11
255.255.255.0
  mac-address aaa.bbb.ccc
```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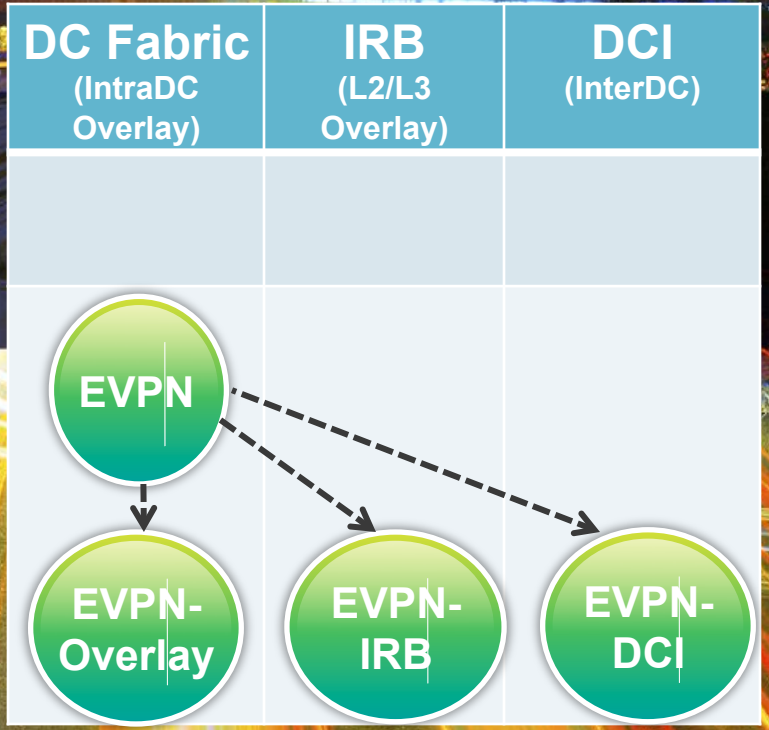
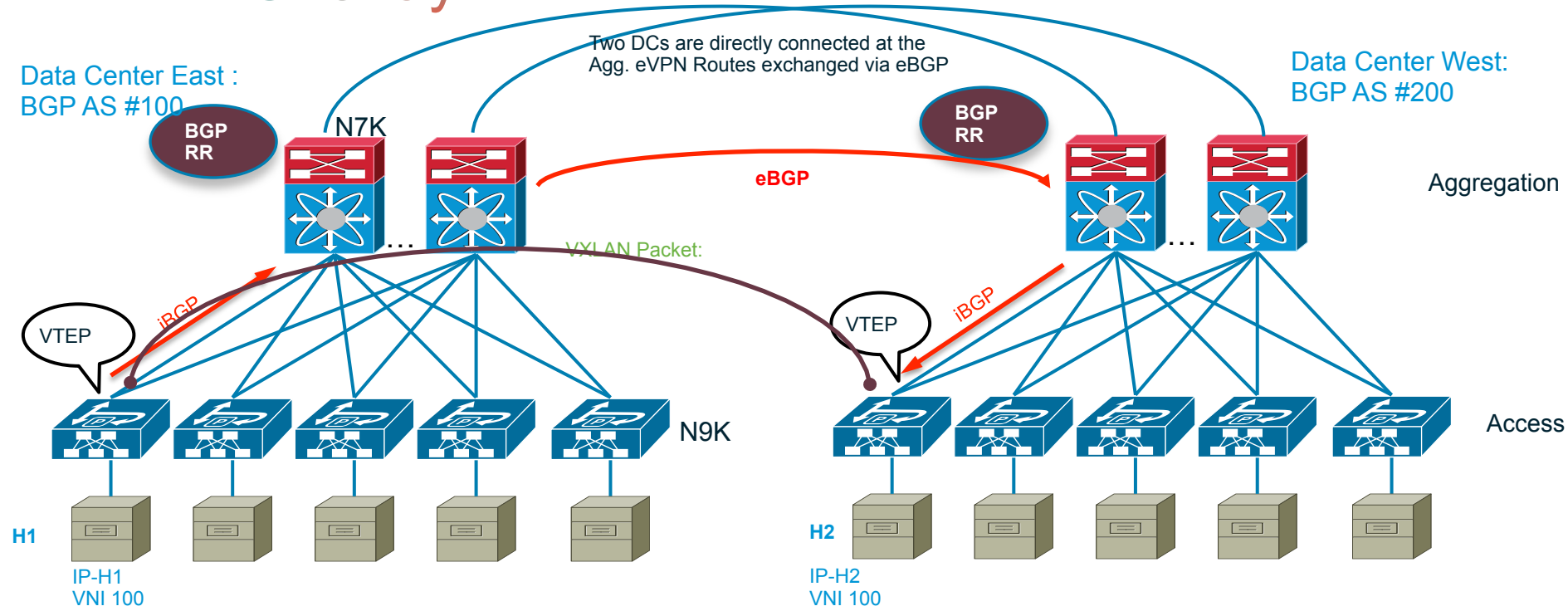
# DC Applications



# EVPN-Overlay




- Layer-2 multi-tenacy
- Geo-redundancy & seamless VM mobility support
- ARP suppression & ARP proxy
- Support large scale VMs & policy control

- Support for different encaps such as VxLAN, NVGRE, MPLS, MPLSoUDP
- Extensive multi-homing support
- Extensive load-balancing capabilities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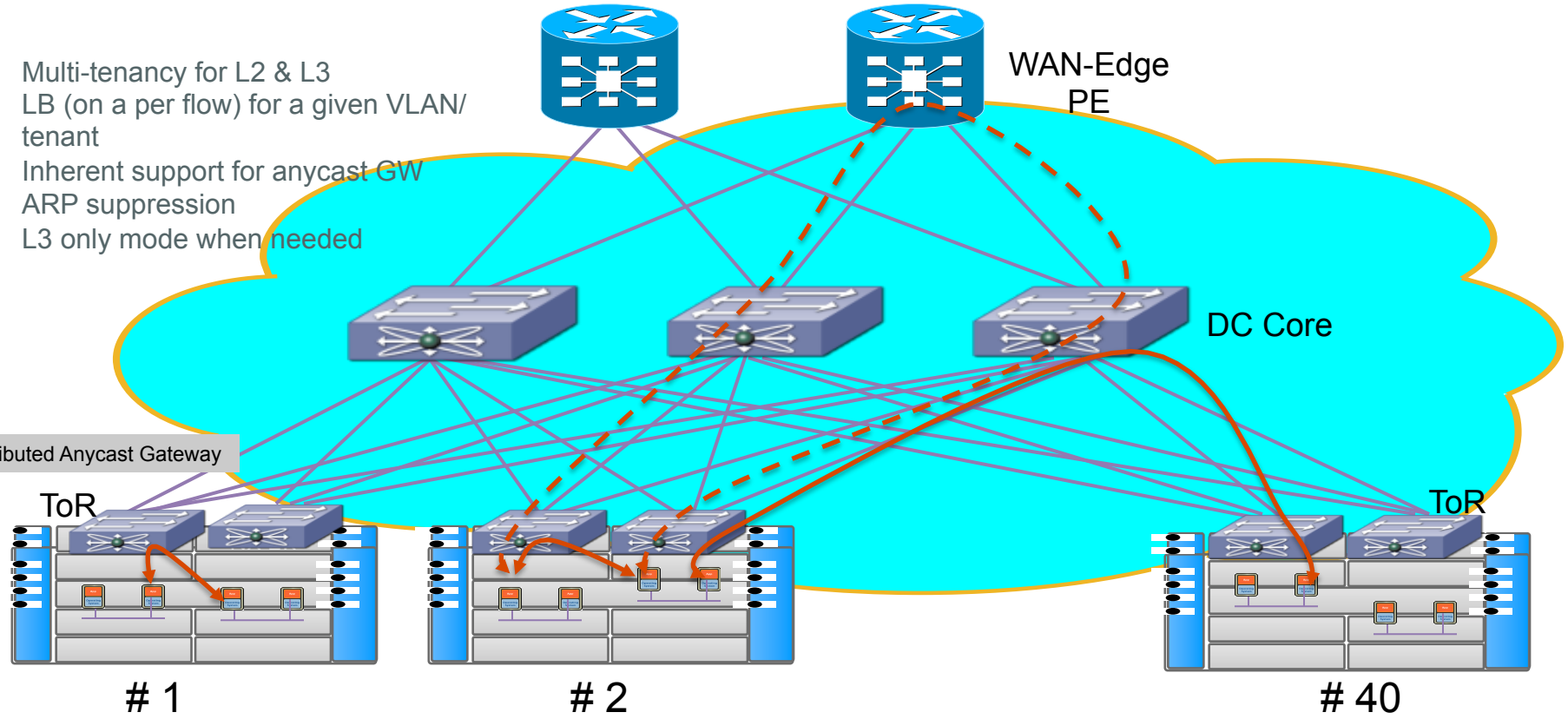
# EVPN-IRB

- Multi-tenancy for L2 & L3
- LB (on a per flow) for a given VLAN/ tenant
- Inherent support for anycast GW
- ARP suppression
- L3 only mode when needed





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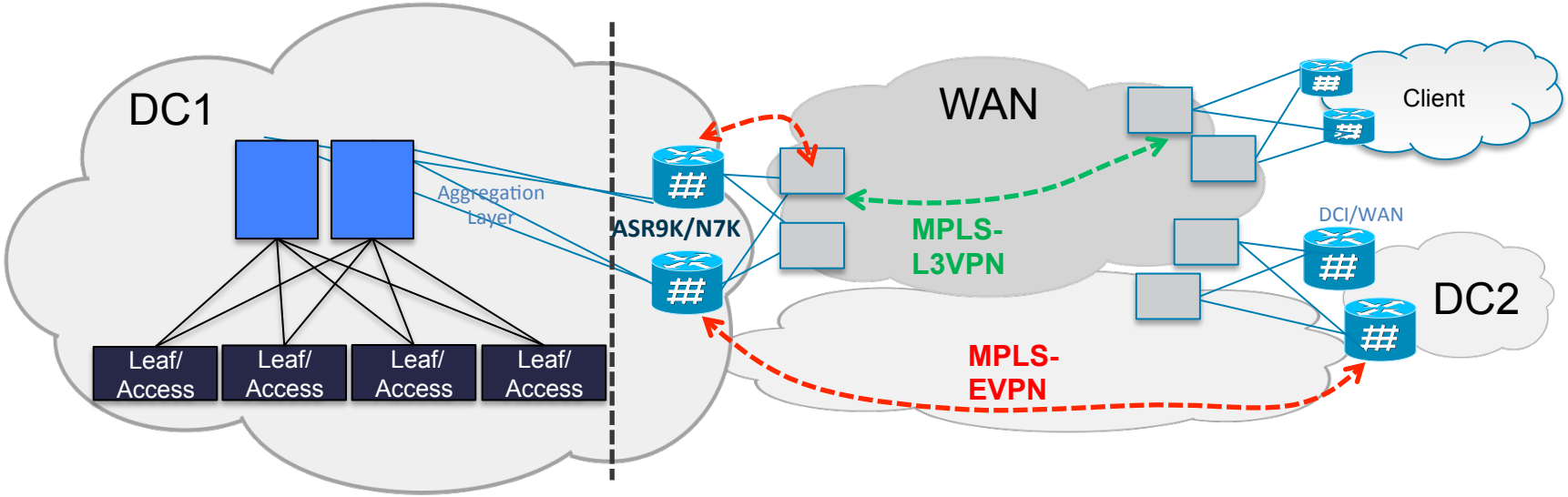
# EVPN-DCI




- Seamless interop with IP-VPN (RFC4364)
- Support for any kind of access technology such as EVPN Overlay (VxLAN), Native Ethernet, TRILL, 802.1Qbp, MPLS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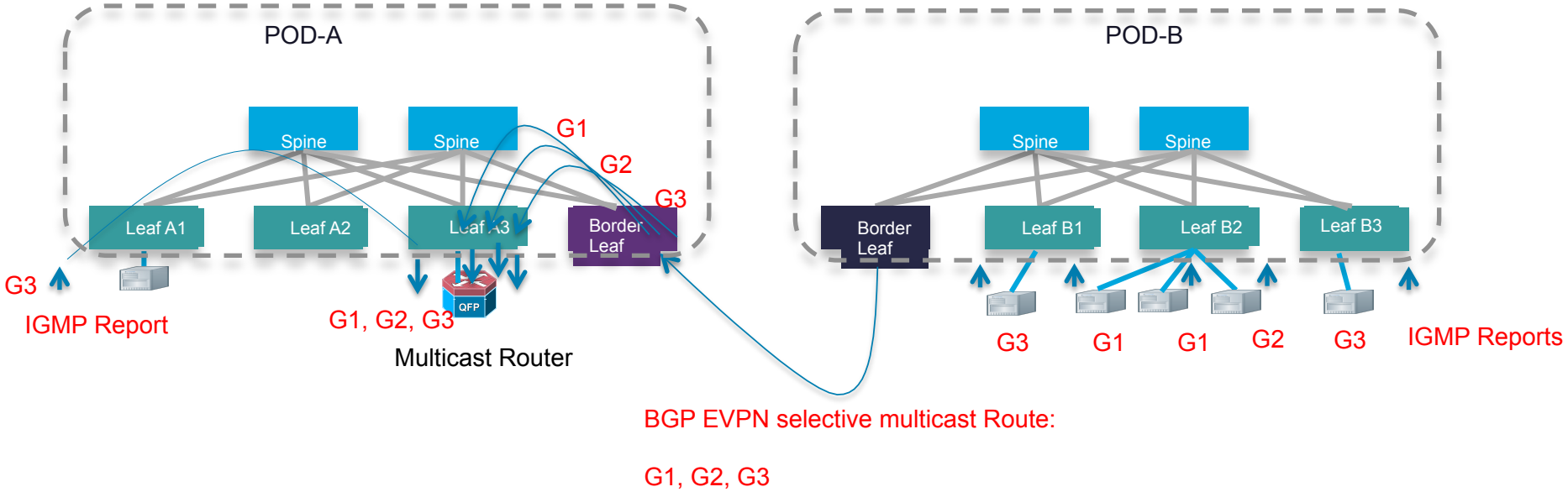
# IGMP/MLD Proxy





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Thank you.